Discrete Mathematics – Final Exam

2024 – Term A

Instructions for students:

1. In the exam there are **8** short questions and **3** open questions.
2. You must answer **all** questions (No choice).
3. The questions are not ordered according to difficulty!
4. Answers should be written on the exam form.
5. The notebooks can be used as drafts but will not be graded.
6. When asked to write explicitly a "set/function/relation," you must specify all the elements of the set/function/relation.
7. Use of any external aiding material is **forbidden**. For your convenience, a formula sheet is attached at the end of the form.
8. The duration of the exam is **three hours.**

Good luck!

Part A – Short Questions

- Write your answers clearly in the marked boxes.
- No need to explain how you got to the final answer, **unless it is explicitly asked**.
- In counting questions, you can use any of the convenient notations which we used in class, **except** for the symbol $\sum$ and three dots, '…'.

Question 1 (6 points)

Given the following propositions, write the numbers of the propositions that are **not** equivalent to:

$$\big(p \to (q \to r)\big) \wedge (s \vee \neg s).$$

1. $\neg(p \wedge q \wedge \neg r)$
2. $(p \vee s) \wedge (q \to \neg r)$
3. $r \vee \neg q \vee \neg p$
4. $(q \to \neg p) \vee r$

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Question 2 (6 Points)

Disprove: If $A \nsubseteq B \cap C$ then $(A \backslash B) \cap (A \setminus C) \neq \emptyset$

$A = \{1\}, B = \{1,2\}, C = \{3\}$

Question 3 (6 points)

Can there exist three sets A, B, C and two functions $f: A \to B$, $g: B \to C$, such that f is 1-to-1 and $g \circ f$ is onto but not 1-to-1?

If the answer is yes, give an example of such sets and functions. If the answer is no, explain <u>shortly</u> why.

Yes. Example:

$$A = \{1,2\}, B = \{3,4,5\}, C = \{6\}$$
$$f: A \to B \quad \{(1,3),(2,4)\}$$
$$g: B \to C \quad \{(3,6),(4,6),(5,6)\}$$

<u>Question 4</u> (6 points)

There are 6 math courses, 6 computer science courses, and 3 physics courses available at the university. Out of the courses offered, what is the number of ways to choose 6 courses including at least one course from each subject?

$$\binom{15}{6} - \binom{9}{6} - \binom{9}{6} - \binom{12}{6} + \binom{6}{6} + \binom{6}{6}$$

(Inclusion-exclusion)

<u>Question 5</u> (6 points)

Let n be a natural odd number. What is the number of ways to spread $5n$ **<u>identical</u>** balls into n bins such that the number of balls in each bin is odd?

$$\binom{3n-1}{2n}$$

Place one ball in each bin (one option).

Pair up the $4n$ remaining balls, to get $2n$ pairs.

Spread the $2n$ pairs into the n bins.

Question 6 (6 points)

What is the coefficient of x^2y^3z in $(x+y)^4(y+z)^2$?

We have 2 terms to choose one occurrence of z from, and 4 terms to choose 2 occurrences of x from.

Given the choices for z and x, the terms from which we take y are uniquely determined.

In total $2 \cdot \binom{4}{2} = 12$

Question 7 (6 points)

Count the number of paths in the plane from (0,0) to (15,20), where each path moves one unit upwards or right with each step, and the path must contain the points (2,3) and (10,10).

$$\binom{5}{2} \cdot \binom{15}{8} \cdot \binom{15}{5}$$

Get from $(0,0)$ to $(2,3)$: 5 steps – 2 right and 3 up. Choose which steps are right: $\binom{5}{2}$

Get from $(2,3)$ to $(10,10)$: 15 steps – 8 right and 7 up. Choose which steps are right: $\binom{15}{8}$

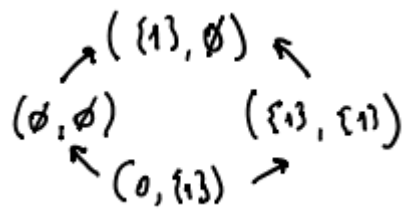
Get from $(10,10)$ to $(15,20)$: 15 steps – 5 right and 10 up. Choose which steps are right: $\binom{15}{5}$

Question 8 (6 points)

Given is a partial order relation R on $P(\{1\}) \times P(\{1\})$, defined as follows:

$$(A,B)R(C,D) \quad \Leftrightarrow \quad (\bar{C} \subseteq \bar{A}) \wedge (D \subseteq B)$$

Draw the Hasse diagram of R, and write the minimum, maximum, and all the minimal and maximal elements (if they exist).



Maximal and maximum: $(\{1\}, \emptyset)$

Minimal and minimum: $(\emptyset, \{1\})$

Part B – Open Questions

- Write your answers clearly in the marked boxes. You must prove your answers.
- **Write clear and readable answers! Your grade will also depend on the clarity of your proof and not only the correctness of your answer!**

Question 9 (16 points)

Prove by induction that every number $n \in \mathbb{N}$ can be written as $n = \sum_{i=0}^{k} a_i\, 2^i$, where $a_i \in \{0,1\}$, and $k \in \mathbb{N}$. For example, $n = 9$ can be written as

$$9 = 1 \cdot 2^0 + 0 \cdot 2^1 + 0 \cdot 2^2 + 1 \cdot 2^3$$

By strong induction on n.

Base case: for $n = 0$ it holds that $n = 0 = 0 \cdot 2^0$.

Let n be a natural number, strictly greater than 0. We split into two cases:

Case 1 – n is even: Then $\frac{n}{2}$ is a natural number strictly smaller than n, so by induction hypothesis we can write $\frac{n}{2} = \sum_{i=0}^{k'} a_i' \cdot 2^i$ where $a_i' \in \{0,1\}$, $k' \in \mathbb{N}$. Hence, $n = 2\sum_{i=0}^{k'} a_i' \cdot 2^i$. Let $k = k' + 1$, and define $a_0 = 0$, and $a_i = a_{i-1}'$ for each $i \in \{1,2,3,\dots,k\}$. Then $n = \sum_{i=0}^{k} a_i \cdot 2^i$, where $a_i \in \{0,1\}$ and $k \in \mathbb{N}$, as needed.

Case 2 – n is odd: Then $n - 1$ is a natural number strictly smaller than n, so by induction hypothesis we can write $n - 1 = \sum_{i=0}^{k'} a_i' \cdot 2^i$ where $a_i' \in \{0,1\}$, $k' \in \mathbb{N}$. Rewriting this we get

$$n - 1 = a_0' + 2\sum_{i=1}^{k'} a_i' \cdot 2^{i-1}$$

So, because $n - 1$ is even, and $2\sum_{i=1}^{k'} a_i' 2^{i-1}$ is even, it must be that a_0 is also even, so $a_0' = 0$.

Thus, for $a_0 = 1$ and $a_i = a_i'$ for $i \in \{1,2,3,\dots,k'\}$ we get

$$n = 1 + 2\sum_{i=1}^{k'} a_i' \cdot 2^{i-1} = a_0 + 2\sum_{i=1}^{k'} a_i \cdot 2^{i-1} = \sum_{i=0}^{k'} a_i \cdot 2^i$$

as needed.

Question 10 (20 points)

Let A, B, C be sets satisfying $(A \times A) \cup (B \times B) = (C \times C)$. Prove the following:

a. $A \cup B \subseteq C$
b. $(B \subseteq A) \vee (A \subseteq B)$

a. Let $x \in A \cup B$. By definition of union, $x \in A$ or $x \in B$. So, by definition of Cartesian product, $(x, x) \in A \times A$ or $(x, x) \in B \times B$. By definition of union, $(x, x) \in (A \times A) \cup (B \times B)$. Since it is given that $(A \times A) \cup (B \times B) = (C \times C)$, we get $(x, x) \in C \times C$. So, by definition of Cartesian product, $x \in C$.
So, we have shown that for every $x \in A \cup B$ it holds that $x \in C$, meaning that $A \cup B \subseteq C$.

b. Assume towards contradiction that $A \nsubseteq B$ and $B \nsubseteq A$ (We used De Morgan). Then there exists $x \in A$ such that $x \notin B$, and there exists $y \in B$ such that $y \notin A$. Since $x \in A$, we have $x \in A \cup B$ (by definition of union), so by item a that we've already proved, we get $x \in C$. Similarly, since $y \in B$, we have $y \in A \cup B$, so by item a we get $y \in C$. Thus, by definition of Cartesian product, $(x, y) \in C \times C$. As it is given that $(A \times A) \cup (B \times B) = (C \times C)$, we obtain $(x, y) \in (A \times A) \cup (B \times B)$. But $(x, y) \notin A \times A$ (as $y \notin A$), and also $(x, y) \notin B \times B$ (as $x \notin B$), therefore $(x, y) \notin (A \times A) \cup (B \times B)$ – contradiction.

Question 11 (16 points)

Let R be a relation on a non-empty set A, and let Q be a relation on A defined as follows:

$$xQy \quad \Leftrightarrow \quad \forall z(xRz \rightarrow yRz)$$

Prove or disprove:

a. Q is reflexive.

b. Q is symmetric.

a. Proof: Let $x \in A$. Then trivially, for every $z \in A$, if xRz then xRz. Hence, xRx.

b. Counter-example: Let $A = \{1,2,3,4\}$ and define $R = \{(2,1), (4,1), (4,3)\}$. It holds that $2Q4$, since if $2Rz$, then it must be that $z = 1$, and hence also $4Rz$. But it doesn't hold that $4Q2$, because for $z = 3$ we have $4Rz$ but $\neg(2Rz)$. Therefore, Q is not symmetric.

An extra page for the open questions:

Discrete Mathematics – Formula sheet for the exam

Set Theory:

$$A\Delta B = (A\backslash B) \cup (B\backslash A)$$
$$A \times B = \{(a,b) |\ a \in A\ and\ b \in B\}$$
$$P(A) = \{S | S \subseteq A\}$$

$$|A \times B| = |A| \cdot |B|$$
$$|P(A)| = 2^{|A|}$$

De-Morgan:

$$\overline{B \cup C} = \bar{B} \cap \bar{C} \qquad \mathrm{A}\backslash(B \cup C) = (A\backslash B) \cap (A\backslash C)$$
$$\overline{B \cap C} = \bar{B} \cup \bar{C} \qquad \mathrm{A}\backslash(B \cap C) = (A\backslash B) \cup (A\backslash C)$$

Distributive:

$$\mathrm{A} \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
$$\mathrm{A} \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Functions/Relations:

$$i_{\mathrm{A}} = \{(a,a) | a \in A\}$$

Logic:

For all formulas α, β, γ :

De-Morgan:

$$\neg(\alpha \wedge \beta) \equiv \neg\alpha \vee \neg\beta$$
$$\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$$

Distributive:

$$\alpha \wedge (\beta \vee \gamma) \equiv (\alpha \wedge \beta) \vee (\alpha \wedge \gamma)$$
$$\alpha \vee (\beta \wedge \gamma) \equiv (\alpha \vee \beta) \wedge (\alpha \vee \gamma)$$

Negation rules:

$$\neg\forall \mathrm{x}(\alpha) \equiv \exists x(\neg\alpha)$$
$$\neg\exists \mathrm{x}(\alpha) \equiv \forall x(\neg\alpha)$$

Combinatorics:

Choosing k items out of n different items:

	With repetition	Without repetition
With order	n^k	$\frac{n!}{(n-k)!}$
Without order	$\binom{k+n-1}{n-1}$	$\binom{n}{k} = \frac{n!}{(n-k)! \cdot k!}$

Binomial Formula

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} \cdot x^k y^{n-k}$$

Multinomial Coefficient:

$$\binom{n}{i_1, i_2, i_3, \dots, i_k} = \frac{n!}{i_1! \cdot i_2! \cdot i_3! \cdot \dots \cdot i_k!} \quad if \;\; i_1 + i_2 + i_3 + \cdots + i_k = n$$

Inclusion-exclusion principle:

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{I \subseteq \{1,2,\dots,n\},\;\; I \neq \emptyset} (-1)^{|I|-1} \cdot \left| \bigcap_{\mathrm{i} \in \mathrm{I}} \mathrm{A_i} \right|$$